

Blue light: What is it and How Does it Affect Our Eyes?

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ABSTRACT

Blue light, a type of high-energy visible (HEV) light with wavelengths ranging from 380 nm to 500 nm, is emitted by digital screens, LED lighting, and sunlight. Its short wavelength makes it more energetic than other colors of visible light. Prolonged exposure to blue light has been associated with digital eye strain and may contribute to retinal damage over time (Harvard Health Publishing, n.d.). Studies indicate that excessive screen time can lead to eye discomfort, sleep disruption, and long-term eye health concerns. Furthermore, blue light exposure, particularly in the evening, is believed to suppress melatonin production—a hormone critical for sleep regulation—potentially resulting in insomnia (Sleep Foundation, n.d.).

To mitigate these effects, managing screen time and utilizing blue light-blocking tools, such as filters or specialized glasses, are recommended. Additionally, adopting good sleep hygiene practices, like limiting screen exposure before bedtime, may support healthier sleep patterns and overall well-being (Harvard Health Publishing, n.d.; Sleep Foundation, n.d.).

DEFINITIONS

High-Energy Visible (HEV) Light

This refers to a type of light within the visible spectrum, but with higher energy compared to other colors. It includes blue light. It's just light that has a lot of energy and is visible to our eyes.

Wavelength (nm)

Wavelength is the distance between two peaks of a wave, usually measured in nanometers (nm). Light has different colors based on its wavelength. Wavelength is how far apart the

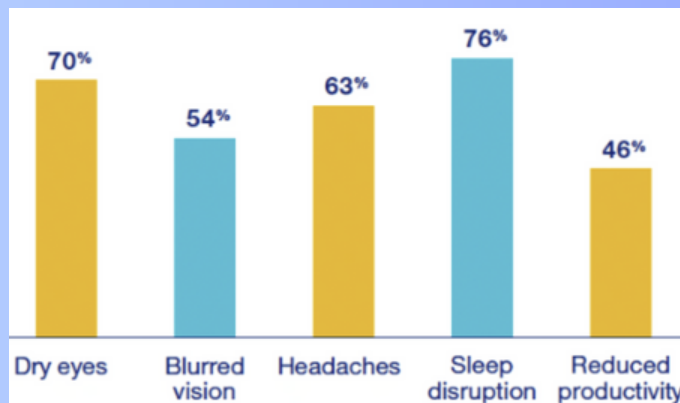


Figure 1: Data from Screen Time Report 2020.

waves of light are. Blue light has shorter waves than red light, which gives it more energy.

Digital Eye Strain

Discomfort that comes from staring at screens for too long, causing headaches, dry eyes, and blurry vision. The tiredness or discomfort in your eyes from too much screen time.

Melatonin

A hormone produced by the brain that helps control your sleep-wake cycle. A natural chemical in your body that helps you sleep.

Screen Time

The amount of time you spend looking at screens (phones, computers, TV). The time spent in front of a screen.

INTRODUCTION

Every day, people spend an average of 7 hours staring at screens, unknowingly exposing their eyes to potentially harmful blue light radiation. Sunlight is the primary source of blue light, but is also found in LED lights and on digital screens. Blue light, a high-energy visible wavelength, plays a critical role in regulating circadian rhythms but has also been implicated

in digital eye strain and sleep disturbances. With screen time surging post-pandemic, understanding the balance between blue light's benefits and risks has become more crucial than ever. It has a wavelength of 380-500 nanometers, making it the shortest wavelength in the visible light spectrum. Despite the growing awareness of its effects, significant gaps remain in understanding its long-term health implications and how best to mitigate them. This review explores these effects, emphasizing the need for future research and practical solutions to safeguard health in a digital era. According to recent reports, the average person spends approximately seven hours daily in front of screens, exposing their eyes to potentially harmful blue light (Nielsen, 2021).

THESIS

Individuals exposed to the blue light from screens, may lead to the disruption in the melatonin production and may increase the eye strain. May suffer from skin issues, disturbances in sleep cycles and other physical and mental health issues. Studies highlight the need for further research into sustainable digital practices.

PROBLEM STATEMENT

Screen time and blue light exposure have been raised as a growing public health concern. Doctors report that symptoms of excessive

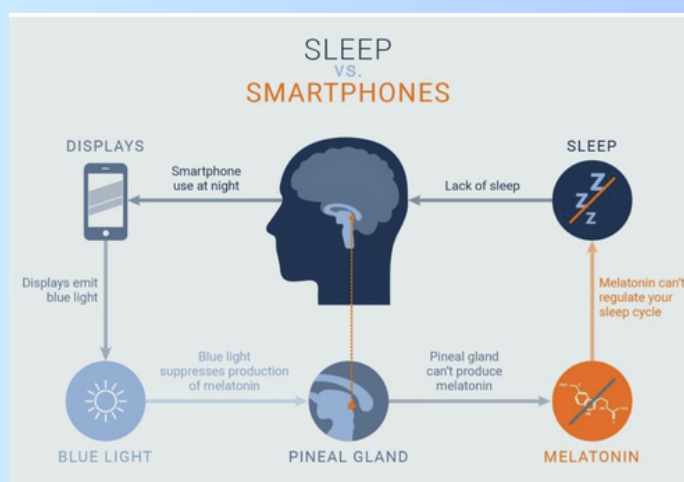


Figure 2: Sleep Vs Smartphones - Louis Nicola, July 2024

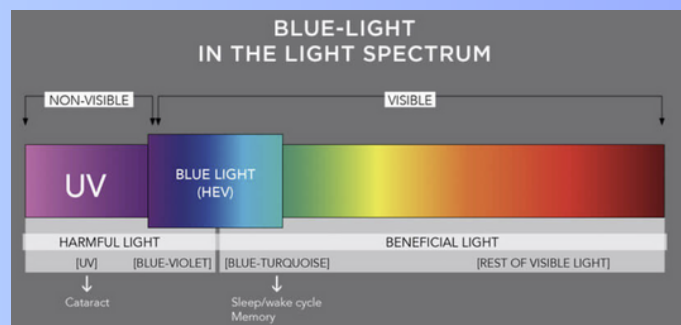


Figure 3: Blue light in light spectrum, Neoretina, January 2024

screen time include sleep disruption (79%), tired or sore eyes (77%), dry eyes (70%), headaches (56%), reduced productivity (46%), and reduced concentration (43%). (Eyesafe, 2020)

HYPOTHESIS

Research done by the vision council highlights that 80% of American adults use digital devices for over two hours daily, with nearly 67% using two or more devices simultaneously. Consequently, 59% report symptoms of digital eye strain, including headaches, blurry vision, and discomfort.

(Cultivating Health, August 2022, The Vision Council, assessed in 2024)

The blue light comes in the visible light bracket ranging from 380 to 500 nm. It can also be sliced into two areas – blue-violet light (about 380 to 450 nm) and blue-turquoise light (around 450 to 500 nm). In essence, as much as one-third of all visible light falls into the High-Energy Visible (HEV) category. In short, blue light. (Neoretina, January 2024)

Also, rays in the red end of the visible light spectrum have less energy due to their longer-wavelengths. And rays in the blue end of the spectrum have more energy by the same logic – because of their shorter-wavelengths. (Neoretina, January 2024)

SYMPTOMS OF DIGITAL BLUE LIGHT

The most common symptoms associated with CVS or digital eyestrain are eyestrain, headaches, blurred vision, dry eyes, neck and

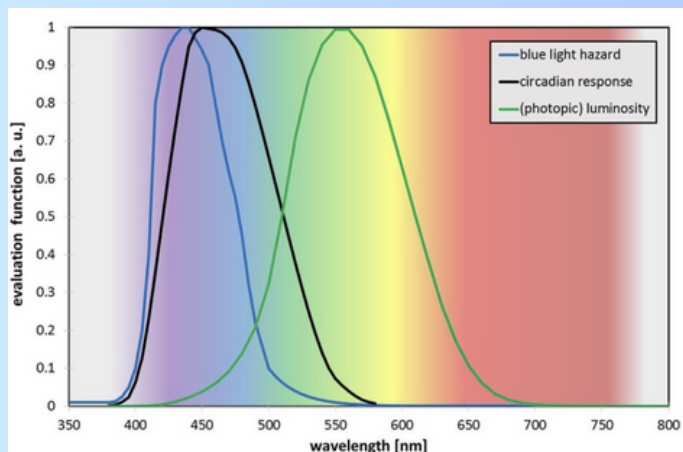


Figure 4: Blue light Hazard, Christian Lingenfelder, 2018

shoulder pain. These symptoms may be caused by poor lighting, glare on a digital screen, improper viewing distances, poor seating posture, uncorrected vision problems, and a combination of these factors. (AOA)

BLUE LIGHT'S DETRIMENTAL EFFECT ON SLEEP

Melatonin, a natural hormone that helps regulate the body's circadian rhythm, typically elevated in the evening hours; however, exposure to artificial blue light (short-wavelength, high-energy visible light) at nighttime can trick the eyes into suppressing melatonin in much the same way natural sunlight does. This suppression is believed to correlate with wakefulness and a delayed or poor sleep cycle. (American Optometric Association, 2017)

Blue light may influence the circadian rhythm. Blue light activates the secretion of melatonin in the pineal gland. This can increase or decrease cortisol expression depending on the time of the day, and thus can affect the circadian rhythm. (Zhao, Z. C., Zhou, Y., Tan, G., & Li, J, 2018)

BLUE LIGHT EFFECT ON HUMAN EYES

The anterior structures of the adult human eye (the cornea and lens) are great at blocking UV rays from reaching the light-sensitive retina at the back of the eyeball. So much so that, less than one percent of UV radiation from the sun can reach the retina – and that is a natural wall, even when a person is not wearing sunglasses. (Neoretina, January 2024)

Blue light emitted from digital devices and the sun can change vital molecules in the eye's retina into cell killers (self destroying molecules/ apoptotic particles). An optical chemistry research conducted at The University of Toledo found that out recently and explained that in context of age-related macular degeneration, and causes of blindness in the United States. (Neoretina, January 2024)

While prolonged, natural UV exposure from direct sunlight can have long - term consequences on vision, the artificial blue light

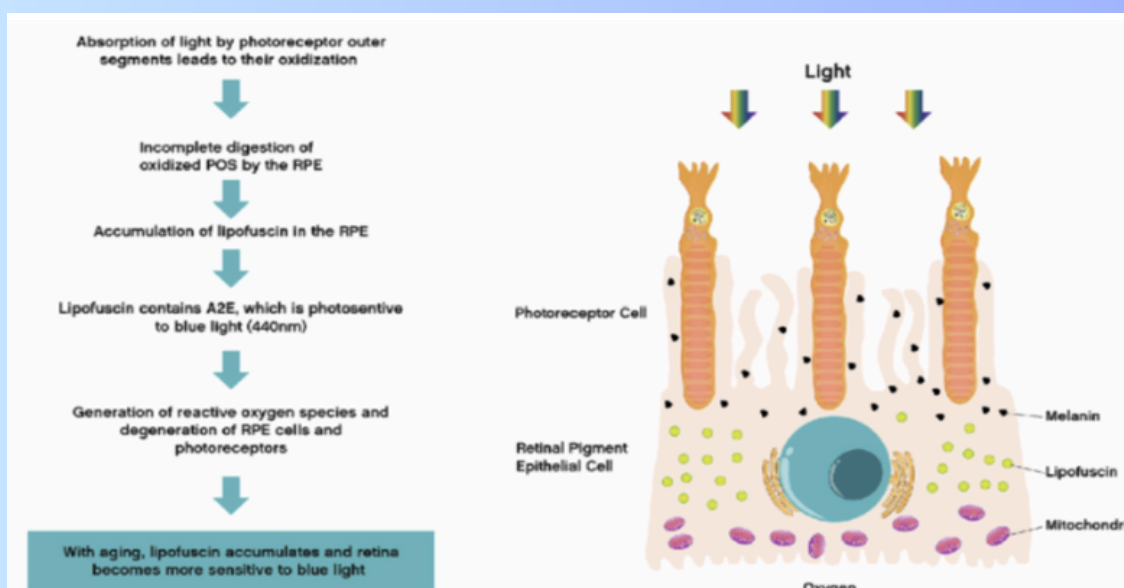


Figure 5: Photochemical damage of the retina , Audrey Cougnard-Gregoire et al., 2023

ROARING CUBS COLLECTIVE

emitted from digital device screens isn't harsh enough to have this effect. That said, extended device use can cause other problems, such as dry eyes, blurred vision or headaches. (American Optometric Association, October 24, 2017)

BLUE LIGHT EFFECT ON HUMAN SKIN

Research shows blue light from electronic devices can lead to changes in your skin cells, including cell shrinkage and death. These speed up the aging process. Even exposures as short as 60 minutes can trigger these changes.

Too much blue light could also lead to changes in skin pigmentation. One study linked blue

20" rule: every 20 minutes, shift your eyes to look at an object at least 20 feet away for at least 20 seconds. You can also use artificial tears to refresh your eyes when they feel dry. (Celia Vimont, Rahul Khurana MD, March 2021)

CONCLUSION

Blue light affects humans in multiple ways. Many adults who use two or more devices at once are more likely to have headaches, eye strain etc. This is because of the blue light that affects. It also affects the sleep cycle of a human being due to the melatonin suppression resulting in delays in sleep. It also damages the human eyes and human skin, by pre aging of skin, or pigmentations. The exposure of the blue

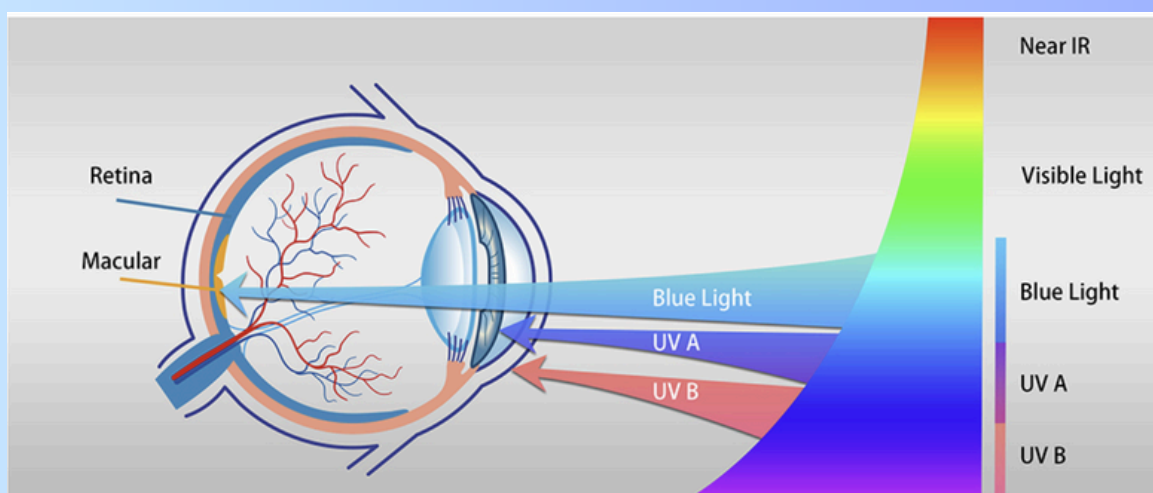


Figure 6: Effect of blue light in retina and contribution in retinal damage, Matias Ko, 29 April 2023

light exposure to more swelling, redness and pigment changes in people with darker skin. (Whitney Seltman, OD, WebMD, 16 September 2022)

A small study published in February 2015 in Oxidative Medicine and Cellular Longevity, found a connection between blue light exposure and the production of free radicals in skin, which have been linked to an accelerated appearance of aging. (Leah Groth, Ross Radusky, MD, 8 November 2021)

REDUCTION OF THE BLUE LIGHT EFFECT

The best way to protect your eyes against eye strain is to take regular breaks using the "20-20-

light may cause pigmentations that are more likely to take place in darker skin people and the redness and swelling of eyes as well increases. To help in the reduction of its causes the best way according to a research from American Academy of Ophthalmology is to use the 20-20-20 rule, relaxing the eyes.